

A Comprehensive Investigation of Shell and Tube Heat Exchangers Based on Porous Baffle Design and Multi-Objective Parameter Optimization

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ABSTRACT

In this paper, the design and performance optimization of porous baffles in shell and tube heat exchanger (STHX) are systematically investigated through numerical simulation and response surface methods. The study focuses on the effects of key parameters such as baffle cut, porosity and number of tubes on the total heat transfer and pressure drop of the STHX.

Keywords: Shell and tube heat exchanger; Porous baffle; Computational fluid dynamics (CFD) ;

1 INTRODUCTION

Shell and tube heat exchangers (STHX) are indispensable key equipment in many industrial fields, including food processing, petrochemicals, and energy production. These new heat exchangers have made significant breakthroughs in heat transfer efficiency, compactness and operational stability, including but not limited to immersion heat exchanger [1], plate heat exchanger [2], spray heat exchanger [3], cased heat exchanger [4], finned tube heat exchanger [5], spiral plate heat exchanger [6], heat pipe heat exchanger [7], microchannel heat exchanger [8] and shell and tube heat exchanger [9] and many other innovative designs, which provide diversified solutions for energy saving and consumption reduction in modern industrial production.

Researchers have worked to achieve significant increases in heat transfer rates through a variety of methods. Numerous studies have focused on enhancing STHX heat transfer. Other approaches include composite reinforcement [17], machine learning optimization [18], and phase change material (PCM) integration [19]. Li et al. [10] optimized baffle design to balance heat transfer and pressure drop. Yilmaz et al. [11] used microchannels and nanofluids to boost convective heat transfer coefficient (CHTC) by 49% (Re~5900). Nie et al.

Current research on porous media in heat exchangers including heat transfer enhancement mechanisms, flow characteristic optimization, and engineering applications still has significant room for in depth exploration. As a key parameter characterizing porous media, porosity plays a critical role: while porous media have been shown to significantly improve STHX heat transfer efficiency, related research remains to be further advanced. Konduru et al. [12] performance of porous media heat exchanger with Kelvin cell structure foam at high temperatures, combined with experimental and numerical simulation party using casting process to integrate the Kelvin foam with cast iron pipe to eliminate the contact thermal resistance.

This paper focuses on STHX performance enhancement by optimizing the design and characterization of porous baffles with the aim of increasing thermal efficiency and reducing pressure drop. The study aims to provide an effective numerical simulation method for the design optimization of shell heat exchangers by means of numerical simulation and response surface analysis. CFD is used to model the STHX equipped with six porous baffles to investigate the temperature and velocity distributions in depth.

2 PROBLEM DESCRIPTION AND RESEARCH METHODOLOGY

2.1 Physical description

Fig. 1 shows three-dimensional cross-sectional views of the STHX with one, four and nine heat exchanger tubes, respectively, along with six equidistant porous baffles. The functionality of the STHX is well known. STHX tube side fluid is gas phase air and shell side fluid is liquid phase water. Between these two, the tube side and shell side fluids exchange heat through the tube wall. The heat exchange area is increased by changing the number of heat exchanger tubes without changing the cross-sectional area of the tube test. Baffles are inserted on the shell side to create turbulence in the liquid phase. The geometric parameters and working conditions necessary for the heat exchanger are also listed in Table 1.

The baffle cut, number of tubes and heat transfer area are key factors affecting the shell flow characteristics, and changes in these parameters can significantly alter the total heat transfer. If the baffles are not installed properly, large-scale recirculation flow and flow separation zone may be formed downstream of it, thus reducing the heat transfer efficiency. Similarly, improperly installed heat exchanger tubes may lead to long-term performance degradation of the STHX or even structural failure. In this paper, the selected baffle cut is 25%, 35% and 50%, the corresponding proportion of the baffle cut area to the cross-sectional area of the shell. At the same time, three working conditions of 1, 4 and 9 tubes were set for the number of tubes, and the comparative analysis of different heat exchanger areas was realized under the condition of constant flow rate of the tubes. Three baffle porosities (fluid volume share) of 20%, 50% and 80% were used to investigate the mechanism of porous media properties on the heat transfer performance of STHX.

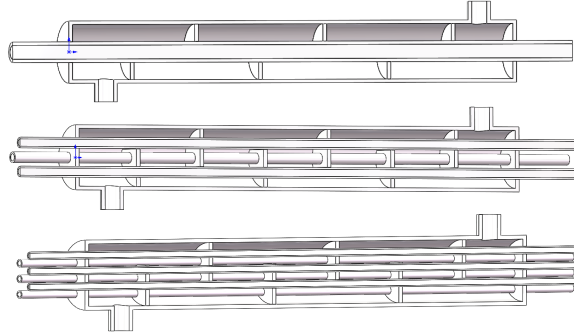


Fig. 1. The cross-section view of the STHX in three dimensions.

Table 1 Parameters related to geometry and boundary conditions

Parameter	Value
Shell diameter (mm)	90
Inner/Outer diameter of tube (mm)	20/30
Tube length (mm)	750
Number of tubes	1/4/9
Inner/Outer STHX length (mm)	590/600
Tube/Shell side velocity flow rate (m/s)	20/1
Tube/Shell side inlet temperature (°C)	200/10
Baffle spacing (mm)	86

2.2 Governing equation

To simplify the numerical simulation while ensuring reliability for the study core objectives, the following modeling assumptions were adopted. The fluid flow and heat transfer processes in the STHX were assumed to be steady-state, where transient fluctuations have minimal impact on long-term performance trends. Incompressible fluid assumption that both tube-side air (gas phase) and shell-side water (liquid phase) were treated as incompressible fluids. For tube-side air, the inlet temperature is 200°C and the operating pressure is near atmospheric pressure (101.3 kPa), where the air density variation is less than 5%, for shell-side water, liquid incompressibility is a standard assumption in low-to-medium pressure heat

exchanger simulations [24]. Porous media simplification: The porous baffles were modeled using the Darcy-Brinkman-Forchheimer model. This assumes uniform porosity distribution in the baffle, which is consistent with the ideal manufacturing state of metal foam porous baffles [15,23]. Negligible radiation heat transfer that only convective and conductive heat transfer were considered, while radiation heat transfer was neglected. Smooth tube wall assumption that the tube inner and outer walls were assumed to be smooth. This simplifies the calculation of wall shear stress, and is a common initial assumption for STHX performance optimization studies [24]. A complete mathematical description of a moving fluid relies on three fundamental physical laws: conservation of mass, conservation of momentum, and conservation of energy. For steady state flow, flow field physical quantities do not change with time this simplifies the expression of governing equations. The set of control equations underlying the isolation solver used in this paper is specifically given by Eqs. (1) to (5), which provide a complete picture of the mass transfer, momentum exchange, and energy transformation processes of the fluid under steady state flow conditions.

$$\text{Conservation of mass: } \nabla \cdot (\rho \vec{V}) = 0 \quad (1)$$

$$\text{x momentum: } \nabla \cdot (\rho u \vec{V}) = -\frac{\partial p}{\partial x} + \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} \quad (2)$$

$$\text{y momentum: } \nabla \cdot (\rho v \vec{V}) = -\frac{\partial p}{\partial y} + \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} + \frac{\partial \tau_{zy}}{\partial z} + \rho g \quad (3)$$

$$\text{z momentum: } \nabla \cdot (\rho w \vec{V}) = -\frac{\partial p}{\partial z} + \frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \tau_{zz}}{\partial z} \quad (4)$$

$$\text{Energy: } \nabla \cdot (\rho e \vec{V}) = -p \nabla \cdot \vec{V} + \nabla \cdot (k \nabla T) + q + \Phi \quad (5)$$

$$\begin{aligned} \Phi &= \mu \left[2 \left[\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial y} \right)^2 + \left(\frac{\partial w}{\partial z} \right)^2 \right] + \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \right)^2 \right] \\ &+ \lambda (\nabla \cdot \vec{V})^2 \end{aligned} \quad (6)$$

where $\vec{V}$ denotes the velocity vector field of the fluid and ρ is the density field of the fluid. The $\vec{V}$ can be decomposed into three directional components in the Cartesian coordinate system, where u , v and w correspond to the velocity components in the x , y and z directions, respectively. In addition, p represents the pressure field on the fluid microelement, τ denotes the shear stress tensor due to viscous interaction, g is the gravitational acceleration vector, and e represents the specific energy per unit mass of the fluid, which includes both internal and kinetic forms of energy.

k represents the thermal conductivity of the fluid, while q represents the rate of heat transfer, which usually includes heat generated by chemical reactions or other heat sources. In this study, there is no volumetric heat source or heat loss in the fluid domain, so q is 0. In addition, Φ is the viscous dissipation function, whose specific expression is given in Eq. 6. λ is the viscosity coefficient.

For the flow problem in porous media, the momentum conservation equations need to be modified by introducing the Darcy-Brinkman-Forkheimer model [13], which has the following expressions in the x , y , and z directions:

$$\frac{\rho}{\varepsilon} \left(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} \right) = -\frac{\partial P}{\partial x} + \mu_{\text{eff}} \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) - \left(\frac{\varepsilon \mu_{\text{eff}}}{\varphi} + \frac{\rho F \varepsilon}{\sqrt{\varphi}} \right) u \quad (7)$$

$$\frac{\rho}{\varepsilon} \left(u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} \right) = -\frac{\partial P}{\partial y} + \mu_{\text{eff}} \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} \right) - \left(\frac{\varepsilon \mu_{\text{eff}}}{\varphi} + \frac{\rho F \varepsilon}{\sqrt{\varphi}} \right) v \quad (8)$$

$$\frac{\rho}{\varepsilon} \left(u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} \right) = -\frac{\partial P}{\partial z} + \mu_{\text{eff}} \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right) - \left(\frac{\varepsilon \mu_{\text{eff}}}{\varphi} + \frac{\rho F \varepsilon}{\sqrt{\varphi}} \right) w \quad (9)$$

where ε is the porosity, μ_{eff} shows the effective dynamic viscosity, φ is the permeability, and F is known as inertia coefficient. φ and F can be calculated by Eq. 10 and Eq. 11.

$$\varphi = \frac{d_p^2 \varepsilon^3}{150(1 - \varepsilon)^2} \quad (10)$$

$$\&F = \frac{1.75}{\sqrt{150}\varepsilon^{1.5}} \quad (11)$$

3 GRID RESOLUTION AND MODEL VALIDATION

3.1 Grid independence test

The verification of grip independence, as a key aspect to ensure the accuracy of numerical simulations, is centred on verifying the convergence properties of the computational solutions and quantifying the stability of the results by systematically optimizing the discretization scheme. This process effectively confirms that the simulation results have reached a mesh-independent state, further encryption of the mesh will not significantly affect key physical quantities (e.g., Nussle number), thus providing quantitative support for the reliability of the simulation conclusions. As shown in Fig. 2 (a), the nussle number shows a monotonically increasing trend with the number of grids. Once the mesh refinement reaches a critical value, the effect of continuing to improve the mesh quality on the simulation results tends to weaken. Considering the computational efficiency and reliability of the results, 1322526 was selected as the final grid number in this study. The final meshing scheme demonstrated in Fig. 2(b) is verified to have completely eliminated the effects of mesh dependence, ensuring the stability and physical realism of the numerical solution.

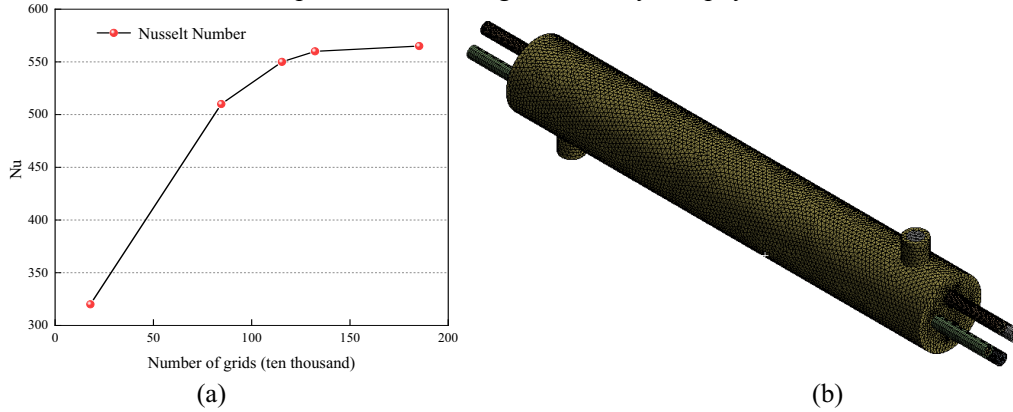


Fig. 2. The results of grid independence test (a) and the final grid solution (b)

3.2 Model validation

This model proposes a new STHX structure with a variable number of porous baffles, which is innovative in terms of design concepts and technological concepts. In view of the lack of experimental studies that are fully consistent with the porous baffle STHX, the study uses CFD simulation data from scholars such as Mohammad Hadi Mohammadi et al. [14] and Ozden [15] on the simple STHX of solid baffles to carry out model validation. As shown in Fig. 3, the comparison of the pressure drop values obtained from the simulation with the literature results shows a minimum error of 6.6% and a maximum error of 13.2%. As the maximum error does not exceed 15%, which is within the acceptable engineering range, the reliability of the calculation results of the simulation model is verified.

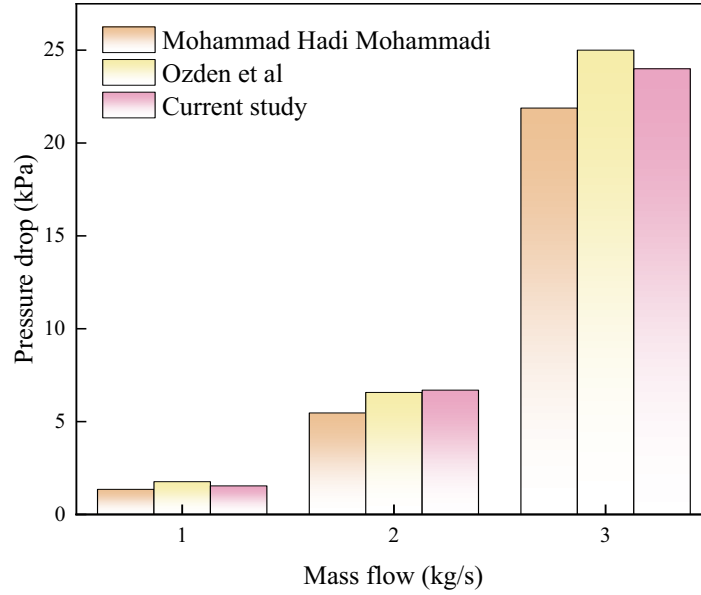


Fig. 3. Validation of the developed simulation model

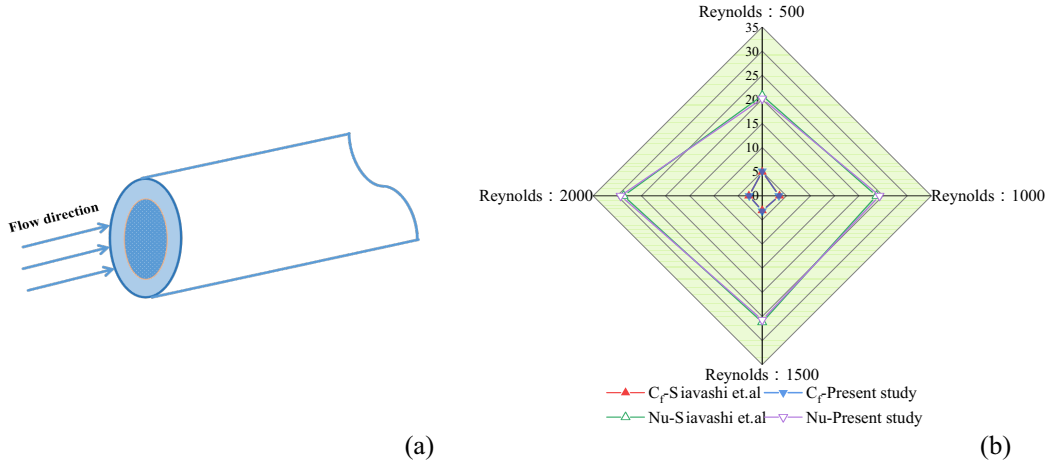


Fig. 4. Porous media model validation. (a) Validation case schematic. (b) Fluid friction factor (C_f) and Nusselt number (Nu) results.

As shown in Fig. 4, the structural schematic diagram and model validation results of the porous media part are presented. Given that pressure drop and heat transfer rate are the two core parameters of concern in this study, the research team selected the fluid friction factor (C_f) and average Nusselt number (Nu) to compare the simulation results of this study with the relevant data from Siavashi et al.[16]. Among them, the C_f can be used as a key parameter reflecting the pressure drop characteristics, and the Nu has a functional relationship with the total heat transfer rate. The comparison results show that the simulation data obtained in this study are highly consistent with the published data by Siavashi et al., which indicates that the currently adopted porous media mathematical model can accurately capture the thermal characteristics of the STHX and provide strong support for the reliability of subsequent research.

4 RESULTS AND DISCUSSION

In this study, a numerical simulation method based on the shear stress transport (SST) $k-\omega$ turbulence model is used to systematically analyze the complex flow characteristics inside a STHX. To improve the numerical stability of the computational process and accelerate the convergence, a pseudo-transient time advancement algorithm is innovatively

introduced in the study. A detailed numerical experimental validation shows that this coupled computational method not only has good numerical stability, but also exhibits excellent accuracy in predicting the STHX experimental observations. The hybrid algorithm is able to capture complex flow phenomena such as flow separation and secondary flow more accurately than the traditional steady-state solution method, providing a reliable numerical analysis tool for heat exchanger performance optimization.

An iterative approach is used to solve the conservation equations (mass, momentum and energy equations) until the computational residuals are reduced to the order of 10^{-7} to ensure the convergence of the numerical solution. In the discretization process, the momentum equations, the turbulent kinetic energy (k) and its dissipation rate (ω), as well as the pressure term are discretized in a second-order windward format. Although the computational convergence of the second-order format is more challenging compared to the first-order discrete format, it has the advantage of significantly improving the accuracy of the numerical simulation, especially in capturing flow details. In addition, the higher-order discrete format effectively reduces numerical diffusion, resulting in a more accurate prediction of the flow and heat transfer characteristics within the heat exchanger.

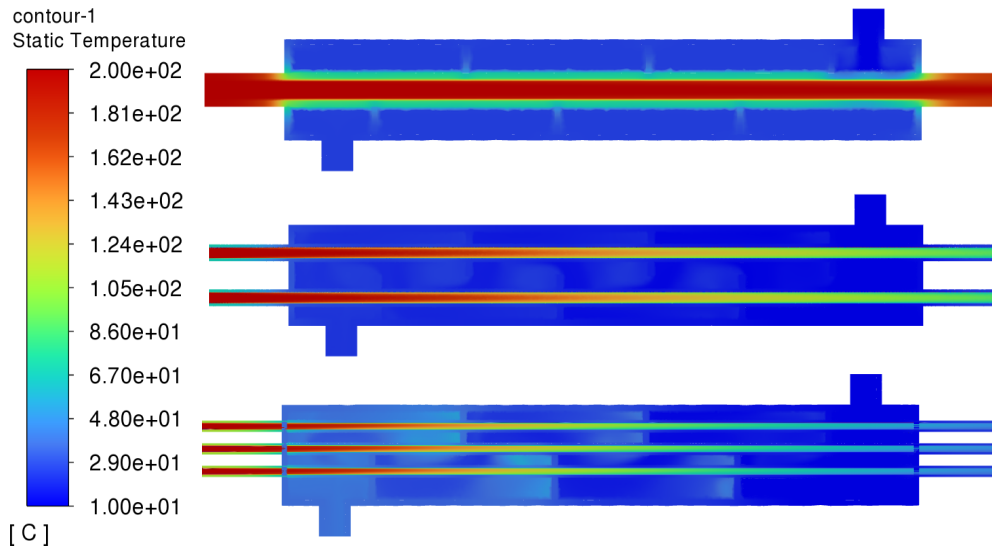


Fig. 7. Temperature distribution of the STHX with the different number of tubes.

4.1 Variations in temperature

Fig. 7 illustrates the temperature distribution characteristics of the STHX under three distinct configurations (1-tube, 4-tube, and 9-tube), which intuitively reflect the heat exchange efficiency differences between the shell-side and tube side fluids. The results show that, driven by the flow-guiding and turbulence-enhancing effects of the porous baffles, the shell side fluid exhibits a continuous and stable decreasing trend in temperature along its axial flow direction (from right to left), this is because the baffles force the shell side fluid to change its flow trajectory periodically, extending the effective contact time between the fluid and the outer tube walls, thereby enabling the fluid to continuously release heat to the tube side fluid through the tube wall and resulting in a gradual temperature drop. In contrast, the tube-side fluid shows a progressive decreasing trend in temperature as it flows through the tube bundle (from left to right), this is attributed to the convective heat transfer between the tube-side fluid and the tube inner walls, as well as the conductive heat transfer through the tube walls. Furthermore, the 9-tube bundle configuration demonstrates superior heat transfer performance, as evidenced by its more pronounced inlet-outlet temperature difference. This enhanced performance can be primarily attributed to improved energy transfer efficiency, reflected in its higher outlet temperature. It should be noted that the current study does not account

for potential local overheating effects, which may significantly influence both temperature field distribution and heat transfer enhancement mechanisms.

4.2 Variations in flow velocity

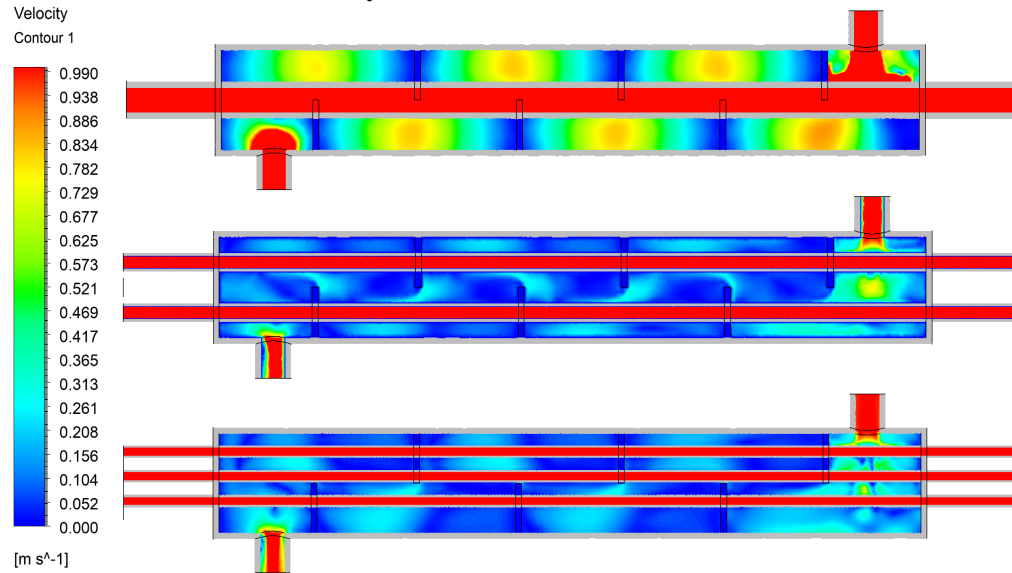
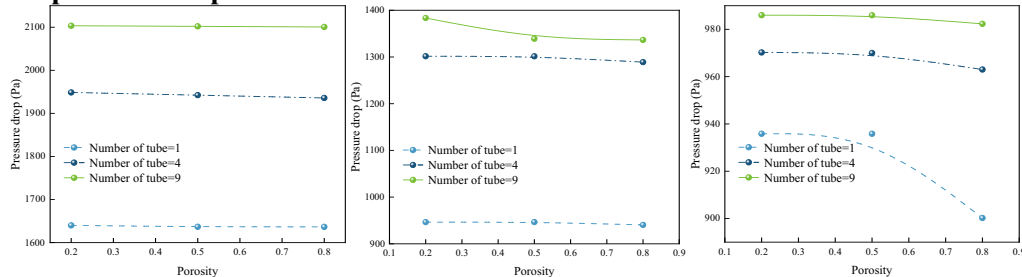


Fig. 8. Velocity (m/s) distribution of the STHX with the different number of tubes.

Fig. 8 illustrates the influence of three typical tube arrangement schemes on the axial velocity field of the STHX, focusing on quantifying the variation of shell side fluid velocity under different tube layout densities. The study results show that when the shell side inlet flow rate is kept constant and the cross sectional area of the tube-side flow channel remains unchanged across all operating conditions, the maximum shell-side flow velocity in the 9-tube arrangement scheme is the lowest specifically, it reaches only 0.82 m/s, which is 31.7% lower than the 1-tube scheme (1.2 m/s) and 18.6% lower than the 4-tube scheme (1.01 m/s). This obvious difference in velocity distribution is mainly attributed to the regulatory mechanism of tube arrangement density on shell side flow resistance: compared with the sparse layout of 1-tube and 4-tube schemes, the 9-tube arrangement has a higher tube density, which makes the shell side fluid need to bypass more tube walls during flow, increasing the frequency of fluid collision and friction with the tube surfaces, at the same time, the narrow flow gaps between densely arranged tubes enhance the local flow constraint effect, leading to a more uniform velocity distribution and a significant reduction in the maximum flow velocity. In the single tube arrangement, the shell flow path is the smoothest, and the fluid only needs to overcome the wall friction resistance. It is particularly noteworthy that in the 9-tube arrangement, the shell flow forms a more complex secondary flow structure in the folding plate region, and the flow velocity gradient in the near-wall region increases significantly, which further aggravates the dissipation of flow energy.

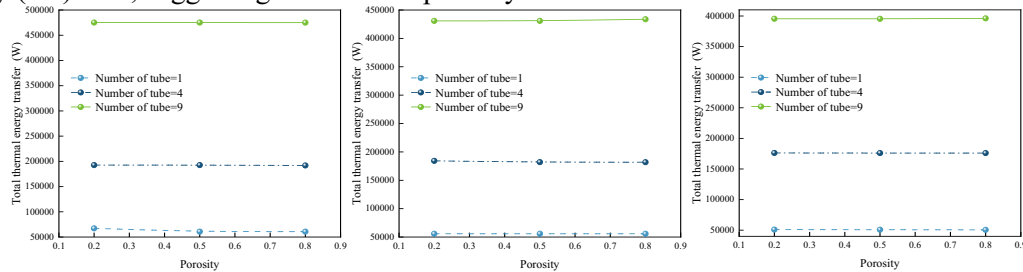
4.3 Effect of tube on pressure drop and heat transfer



(a)Baffle cut=25% (b)Baffle cut=35% (c)Baffle cut=50%

Fig. 9. Pressure drop variation with heat exchanger tubes at different baffle cut (a) 25%, (b) 35%, and (c) 50%

As shown in Fig. 9, the law of influence of the number of heat exchanger tubes (1, 4, and 9 with constant cross-sectional area) on the shell-side pressure drop is simulated for different baffle cut and porosities. The results show that pressure drop decreases as porosity increases. This decrease is more significant for multi-tube bundles (4/9 tubes) than for single tubes. This indicates that high-porosity baffles have a better drag reduction effect on complex flow fields. Simultaneously, the pressure drop exhibits a distinct stepwise escalation with increasing tube count, demonstrating that the influence of tube quantity predominates over porosity regulation in determining the overall pressure drop characteristics. It is noteworthy that the decreasing rate of pressure drop in the multitube bundles tends to slow down at high porosity, which may be due to the existence of a critical porosity threshold, beyond which there is a marginal decrease in the drag reduction benefit. In addition, the magnitude of pressure rise due to the increase in the number of tubes at low porosity (0.2) is significantly higher than that of the high porosity (0.8) case, suggesting that the low porosity flow field is more sensitive to tube bundle perturbations.

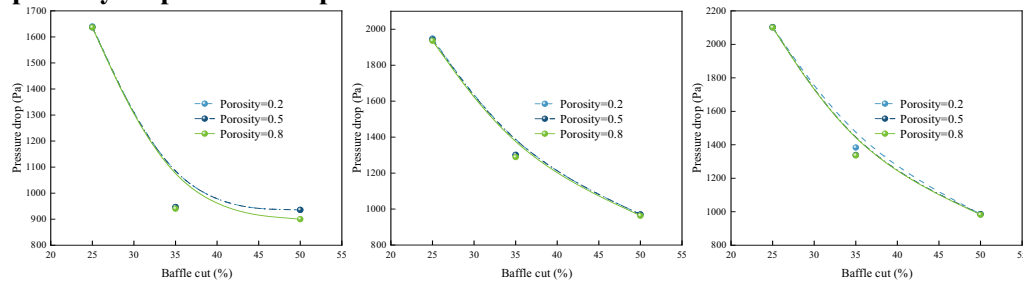


(a)Baffle cut=25% (b)Baffle cut=35% (c)Baffle cut=50%

Fig. 10. Total thermal transfer variation with heat exchanger tubes at different baffle cut (a) 25%, (b) 35%, and (c) 50%

As shown in Fig. 10, the increase in the number of tubes has a decisive effect on the total heat transfer volume at constant cross-sectional area. When the number of tubes increases from 1 to 9, the system heat exchange capacity shows a significant non-linear increase. At a porosity of 0.2 and a baffle cut of 25%, the total heat transfer increased from 67,245.6 W to 475,113.3 W, an increase of 186.4%. Similarly, 182.0 % and 182.4 % enhancements were obtained at porosities of 0.5 and 0.8, respectively. It is worth noting that this enhancement effect shows a clear phasing: when the number of tubes is increased from 1 to 4, 65.8% to 68.9% of the total enhancement effect is achieved, whereas when the number of tubes is increased from 4 to 9, it results in only 31.1% to 34.2% of the additional gain. Particularly importantly, the intensification ratios for different porosity conditions (0.2-0.8) remained highly consistent ($186.4\% \pm 4.4\%$). For any fixed tube number, the difference in total heat transfer between the maximum porosity (0.8) and minimum porosity (0.2) is less than 5%. In contrast, increasing the number of tubes from 1 to 9 leads to an over 180% increase in total heat transfer. This direct data comparison clearly shows that the heat transfer enhancement driven by tube number is far more significant than the slight regulatory effect of porosity, which fully demonstrates that the effect of the number of tubes significantly outweighs the moderating effect of the porosity in terms of total heat exchange regulation.

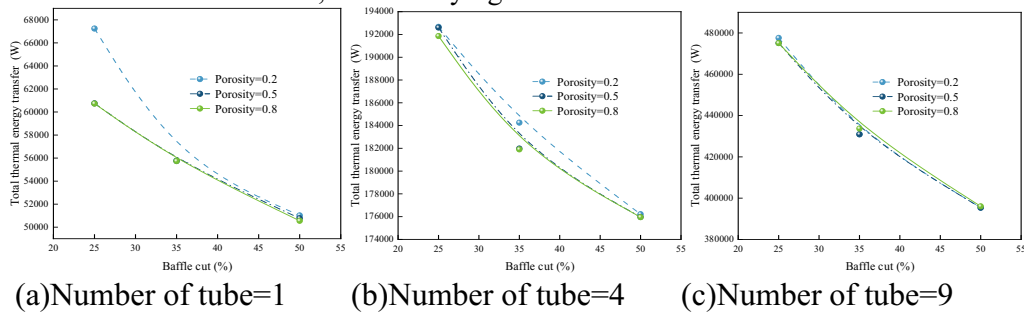
1.1 4.4 Effect of porosity on pressure drop and heat transfer



(a)Number of tube=1 (b)Number of tube=4 (c)Number of tube=9

Fig. 11. Pressure drop variation with Porosity at different number of tube (a) 1, (b) 4, and (c) 9

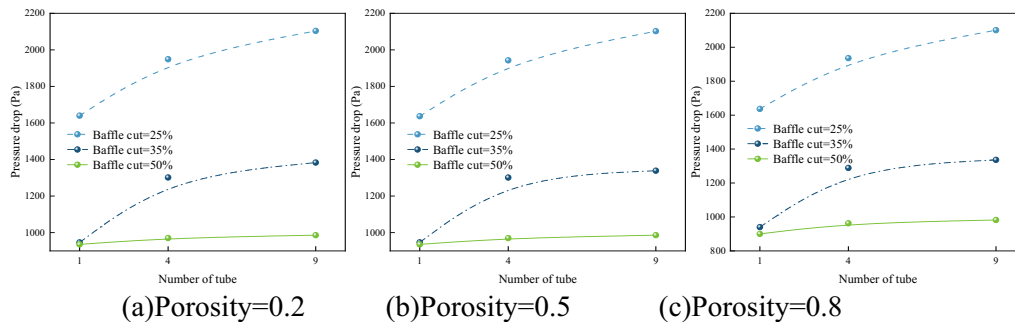
As shown in Fig. 11, the law of porosity effect on shell side pressure drop is systematically analyzed for different number of heat exchanger tubes with baffle cut (0.25, 0.35, 0.5). The results show that the pressure drop decreases significantly with increasing baffle cut, with a maximum drop of 42.9% (1640.1 to 935.8 Pa) in the single-tube system, and more than 50% in the multi-tube system (4/9 tubes). Notably, the effect of porosity is highly dependent on the operating conditions. The difference in pressure drop across various porosities is less than 0.8% at a low baffle cut of 0.25, but widens significantly to 3.8% at a high baffle cut of 0.5. Furthermore, the underlying mechanism shifts near a critical baffle cut of 0.35.



(a) Number of tube=1 (b) Number of tube=4 (c) Number of tube=9
Fig. 12. Total thermal transfer variation with Porosity at different number of tube (a) 1, (b) 4, and (c) 9

As shown in Fig. 12, the effect of porosity on the total heat transfer shows a significant working condition dependency. In single-pipe system, the difference in total heat transfer corresponding to different porosities decreases from 10.1% to 0.9 % when the baffle cut is increased from 0.25 to 0.5, suggesting that the porosity effect is significantly weakened by high baffle cut. The analysis reveals two key findings regarding porosity's role: in the 4-tube system, a medium baffle cut (0.35) defines an optimal porosity range, yielding a marginal (0.1%) yet distinct total heat transfer improvement through parameter synergy. Conversely, porosity exhibits the weakest effect (<0.2%) in the 9-tube system, confirming that high tube density diminishes its regulatory influence.

4.5 Effect of baffle cut on pressure drop and heat transfer.



(a) Porosity=0.2 (b) Porosity=0.5 (c) Porosity=0.8
Fig. 13. Pressure drop variation with baffle cut at different porosity (a) 0.2, (b) 0.5, and (c) 0.8

As shown in Fig. 13, the pressure drop exhibits a significant non-linear decreasing pattern with increasing baffle cut, with a maximum drop of up to 50.2% (4-pipe system), and this effect stems from the expansion of the flow channel cross-section and the reduction of the flow velocity induced by the change in baffle cut. The baffle cut regulates the pressure drop through three distinct mechanisms: it alters the cross-sectional area of the main flow channel, modulates the intensity of flow disturbance induced by the tube bundle, and governs the proportion of flow that leaks through the pores.

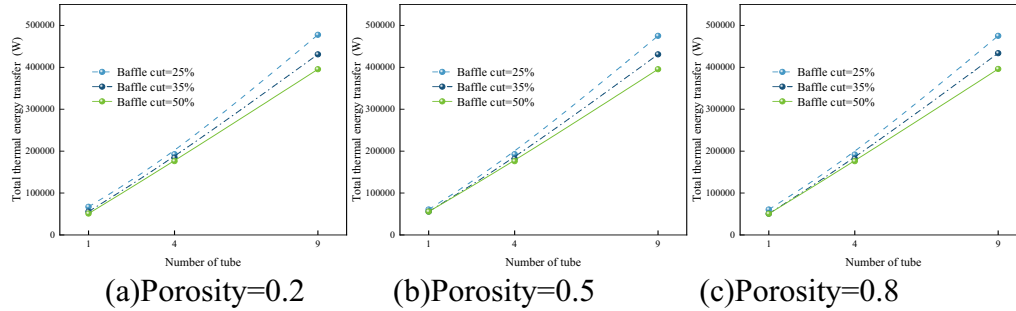


Fig. 14. Total thermal transfer variation with baffle cut at different porosity (a) 0.2, (b) 0.5, and (c) 0.8

As shown in Fig. 14, in the 1-tube system, the heat exchange presents a maximum 24.1% decrease with increasing cut rate (67245.6 to 51017.8 W). In contrast, the 4 and 9-tube systems exhibit 8.5% and 17.2% degradation, respectively, and this discrepancy stems from the essential difference in the flow field characteristics at different tube bundle densities. It is worth noting that there is a clear parametric coupling characteristic of the baffle cut effect. A significant response difference of 6.9% was observed between the 1-tube and 9-tube systems at low porosity (0.2). However, this difference exhibited a dramatic reduction to below 1% under high porosity conditions, highlighting its role in diminishing system-dependent variability.

6 CONCLUSIONS

In this study, STHX with a porous baffle structure is investigated to improve heat transfer performance. After completing the model validation, this study deeply analyses the influence law of parameters such as baffle cut, number of tubes and porosity, and carries out the agent model construction and multi-objective optimization design on the basis of this study. The main conclusions are as follows:

- Through numerical simulation and response surface analysis, it is found that among the baffle cut, porosity and the number of tubes, the folding plate cutting rate is the dominant factor with 89.5% influence on the pressure drop. The number of tubes contributes up to 84.5% to the total heat transfer, which is the decisive parameter for the heat transfer performance.

- Based on the polynomial regression model constructed by the response surface method, combined with the entropy weight TOPSIS decision-making method, the optimal design parameter combinations were determined as the number of tubes 9, porosity 0.705, and baffle cut 0.447. This combination significantly reduces the pressure drop while maintaining high heat transfer efficiency, verifying the effectiveness of the porous folded plate structure in balancing heat transfer and flow resistance in shell-and-tube heat exchanger.

CREDIT AUTHORSHIP CONTRIBUTION STATEMENT

Bo Tian: Conceptualization, Methodology, Software, Investigation, Validation, Visualization, Formal analysis, Writing – original draft. **Ningbo Wang:** Conceptualization, Review & editing, Supervision. **Lu Liu:** Review & editing. **Peng peng:** Data curation. **Yanhua Guo:** Conceptualization. **Li Liu:** Conceptualization. **Shuangquan Shao:** Conceptualization, Methodology, Supervision, Funding acquisition, Writing – review & editing.

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DECLARATION OF COMPETING INTEREST

On behalf of all authors, the corresponding author states that there is no conflict of interest.

DATA AVAILABILITY

Data will be made available on request. All developed codes and models supporting the findings of this study are available from the authors upon reasonable request.

REFERENCE

- [1] Liu Z, Sun X, Ji S, Yuan K. A multi-objective optimization of a single-phase immersion cooling system at cabinet level. *Energy and Buildings*. 2024;312.
- [2] S V, S A, A M. Experimental insights into heat transfer and pressure drop: A comparative study of copper spray coated and plain corrugated gasket plate heat exchangers. *Thermal Science and Engineering Progress*. 2024;56.
- [3] Shim J, Ki S, Seo D, Moon B, Bang S, Nam Y. Intermittent spray cooling on rationally-designed hierarchical surfaces for enhanced evaporative heat transfer performance. *International Communications in Heat and Mass Transfer*. 2024;153.
- [4] Ji J, Pan Y, Zhang J, Zhao D, Chen Q. Optimizing Baffle Height in Improved Elastic Tube Bundle Heat Exchangers. *Journal of Thermophysics and Heat Transfer*. 2024;1-10.
- [5] Cao J, Yuan Y, Zhang Z, Xiao Z, Wang X. Numerical investigation and software design for internally finned helically coiled tubes heat exchangers with single-stream and multi-stream. *Case Studies in Thermal Engineering*. 2024;62.
- [6] Rangasamy R. Experimental and numerical studies of a spiral plate heat exchanger. *Thermal Science*. 2014;18:1355-60.
- [7] Mayoof OT, Yasin NJ, Abedalh AS. Experimental investigation for utilization of U-shaped heat pipe heat exchanger in the air-conditioning system. *International Communications in Heat and Mass Transfer*. 2025;163.
- [8] Jin Q, Chen X, Yang C, Bao J, Zheng J. ANN-based optimization of disk-shaped microchannel heat exchanger for thermal and hydraulic performance improvement. *International Journal of Thermal Sciences*. 2025;213.
- [9] Zhang B, Yang Y, Wu Y. Study on the performance of a novel sinusoidal staggered shell and tube heat exchanger without baffle with experiment verification and CFD modeling. *Journal of Energy Storage*. 2025;114.
- [10] Li N, Chen J, Cheng T, Klemeš JJ, Varbanov PS, Wang Q, et al. Analysing thermal-hydraulic performance and energy efficiency of shell-and-tube heat exchangers with longitudinal flow based on experiment and numerical simulation. *Energy*. 2020;202.
- [11] Yılmaz MS, Ünverdi M, Küçük H, Akcakale N, Halıcı F. Enhancement of heat transfer in shell and tube heat exchanger using mini-channels and nanofluids: An experimental study. *International Journal of Thermal Sciences*. 2022;179.
- [12] Konduru RN, Farges O, Schick V, Hairy P, Gaillard Y, Parent G. Experimental and numerical investigation of porous heat exchangers with Kelvin cell structured foam at high temperatures: Coupled conduction-convection and radiation heat transfer. *International Journal of Heat and Mass Transfer*. 2024;224.
- [13] Pourrahmani H, Siavashi M, Moghimi M. Design optimization and thermal management of the PEMFC using artificial neural networks. *Energy*. 2019;182:443-59.
- [14] Mohammadi MH, Abbasi HR, Yavarinasab A, Pourrahmani H. Thermal optimization of shell and tube heat exchanger using porous baffles. *Applied Thermal Engineering*. 2020;170.
- [15] Ozden E, Tari I. Shell side CFD analysis of a small shell-and-tube heat exchanger. *Energy Conversion and Management*. 2010;51:1004-14.
- [16] Siavashi M, Talesh Bahrami HR, Aminian E. Optimization of heat transfer enhancement and pumping power of a heat exchanger tube using nanofluid with gradient and multi-layered porous foams. *Applied Thermal Engineering*. 2018;138:465-74.